

An Exploration of Natural Batter Swing Tendencies

CSAS 2025 Data Challenge

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Abstract

The 2024 MLB Season introduced Statcast bat-tracking technology that provided new data on bat speed and swing length. While there has been some published work on these topics using preliminary data from early in the season, this is still a developing field with significant amounts of ongoing research. Using data provided by the CSAS 2025 Data Challenge that spans the entire season, this project completes an exploratory data analysis with R to understand the batter tendencies that are reflected in bat speed and swing length, as well as how these are influenced by various situational and pitching variables. Additionally, the relationship between these tendencies and on-field success was studied. Batters were found to maintain a consistent approach under most circumstances, and adjustments primarily occurred due to pitch type, pitch location, and count. Bat speed was also identified as an important predictor for success, but swing length was less closely related. While largely unsurprising, these results were still interesting and serve as the first step towards a larger project. Potential applications for further work include integrating this with biomechanical data or bat tracking data from future seasons to develop new quantitative models for optimizing batter performance.

Introduction

Newly implemented Statcast bat-tracking technology allowed bat speed and swing length data to be collected for the first time during the 2024 MLB Season. Some high-level analyses on this data were published along with its release in May 2024 (Passan, 2024; Petriello, 2024), but this is still a novel topic in baseball lacking extensive study. Bat speed has long been considered using an intuitive approach: A faster swing contains more energy to transfer into a batted ball, and thus players have typically sought to maximize success by swinging hard (Petriello, 2024). This new data allows for more quantitative analysis on this topic, much like how pitch-tracking data led to significant research concerning spin rate and velocity for pitchers (Passan, 2024). This project seeks to develop a thorough understanding of how bat speed and swing length tendencies change based on different situational factors. By identifying how these tendencies relate to on-field success, this serves as the first step towards developing more complex models to optimize batter outcomes. The results could also be expanded with data from future seasons to understand where adjustments are occurring at both the MLB and team levels.

Background

This data for bat speed and swing length adds a completely new dimension to the study of hitting in baseball. Batters have always understood the general physics behind this, but players and researchers were interested in quantifying those notions (Petriello, 2024), continuing the growth of baseball as a sport driven by analytics. As with any new data, bat speed and swing length could present new avenues for batters to make adjustments and improve their performance. Such research into developing models directly linking these measurements to on-field success is likely occurring within analytics departments across the MLB during this offseason. However, this project focuses on simply understanding the data thoroughly, a critical first step to inform any predictive models.

Data from the first month of the 2024 MLB season was initially released in May, and ESPN and MLB.com both published articles analyzing this initial data (Passan, 2024; Petriello, 2024). Most of those findings were unsurprising considering conventional wisdom, but it was insightful to finally have this quantitative evidence. For example, Giancarlo Stanton is known for his leading exit velocity metrics, and this was similarly reflected in bat speed (Passan, 2024). This project expands many of those ideas to data from the entire 2024 season for a much larger sample size, and it considers the relationships between bat speed and swing length with many more variables. Additionally, this project takes a broader approach to understanding MLB batters as a whole, largely ignoring comparisons of individual players.

The initial results from the articles were largely expected but nonetheless interesting. While bat speed is certainly not a deterministic factor, there is evidence for its importance: Swings with 75+ mph bat speed significantly outperformed swings below 75 mph in metrics such as batting average, slugging, and wOBA (Petriello, 2024). Additionally, shorter swing lengths were found to support a more contact-oriented approach in batters such as Luis Arraez and Steven Kwan, while longer swing lengths contributed to more strikeouts for batters such as Stanton and Javier Baez. This was also supported by the batting average, slugging, and whiff rate metrics (Petriello, 2024). Passan (2024) applied these considerations to analyze the performance of many players based on their relative bat speed and swing

length, including explanations for how Juan Soto is so successful while Nolan Arenado has struggled offensively. Understanding this data at the aggregate level can further reveal improvements in batter tendencies with respect to other potential factors, but this could also present another tool for pitchers to study and combat batter adjustments. Regardless of the application, further analysis of this data will have major implications for baseball.

Methods

The raw dataset provided by the CSAS 2025 Data Challenge was difficult to analyze due to its sheer size, but it was otherwise relatively tidy. Using tools from the `tidyverse` in R, the dataset was first explored to select all potentially relevant variables, recoding many of them to be better suited for analysis. Two clean datasets were then extracted from the raw data: One preserved all observations of a swing (with recorded bat speed and swing length), and the other preserved all pitches to determine pitcher tendencies. Table 1 gives a breakdown on how these three datasets compare with each other.

Table 1: Comparisons Between the Raw and Cleaned Datasets

	Raw Data	Swing Data	Pitch Data
Observations	701,557	322,556	701,557
Variables	113	42	31
Character	16	3	3
Date	1	1	1
Factor	0	13	11
Logical	8 (deprecated)	3	3
Numeric	88	22	13

Exploratory data analysis was systematically performed on this clean data, focusing primarily on the swing dataset. The univariate distributions of bat speed and swing length were compared with results from the initial articles. Batter tendencies were then explored in smaller subsets across variables that were hypothesized to cause potential adjustments. These tendencies were related to success on two levels: for individual batters with at least 500 recorded swings, aggregating bat speed and swing length from the entire dataset, and for all swing observations. The batter summary computed median and 95th percentile bat speeds, ignoring outliers of check swings and bunts much like the Statcast seasonal average (Petriello, 2024). The `GGally` and `corrplot` packages were used to visualize potential correlations between bat speed and swing length with every other variable at the swing level, and the strongest correlations were further analyzed. Pitching tendencies were analyzed less rigorously, focusing on areas where pitching-related factors were important in dictating batter tendencies. This was approached from the perspective of understanding how the batter-pitcher interaction could explain the observations. With this project focusing on analyzing batter tendencies, some less relevant pitching trends were not explored.

Results

As a whole, the results were interesting but largely expected. In hindsight, this is unsurprising considering the history of baseball being played for over 100 years, and that baseball data has been analyzed extensively for the past couple of decades. Even in an area that has not been studied directly, player tendencies should be near-optimal since players must be successful to remain at the MLB level. Nonetheless, the alignment of these results with expectations is still notable.

Batter Tendencies

It is important to first understand basic batter tendencies before any more complex analysis. Figure 1 shows the distribution of bat speed and swing length across the entire dataset. Most of the data exhibits a roughly symmetric and unimodal distribution in one region (50-90 mph for bat speed, 5-10 feet for swing length), reflecting the expected variance within a population of players, and the smaller observations likely correspond to events such as check swings and bunts. Bat speed is centered around a median of 70.9 mph and swing length has a median of 7.3 feet, matching closely with the average bat speed of 71.5 mph and average swing length of 7.3 feet from the initial analysis (Passan, 2024).

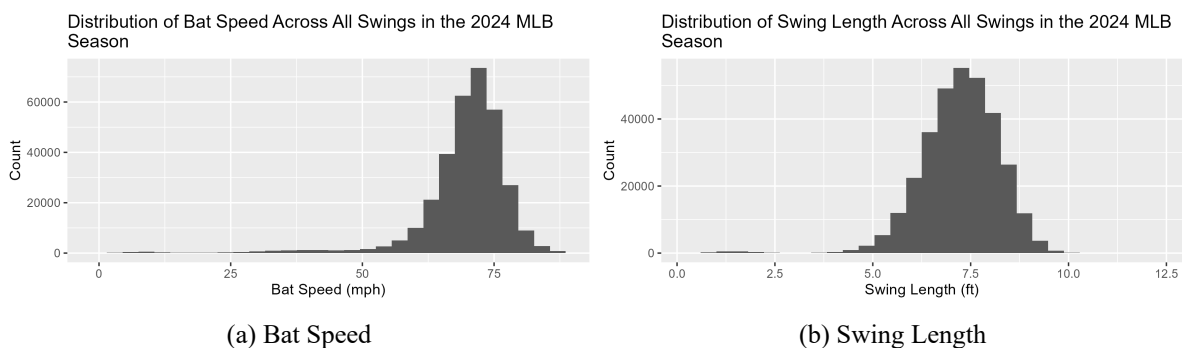


Figure 1: The distributions of bat speed and swing length across all swings in the 2024 MLB Season are unimodal. When considering only full swings, the distributions are roughly symmetric.

Batters were found to seldom adjust their approach, with similar distributions of bat speed and swing length among most situational variables. For example, the platoon advantage illustrates how handedness plays a major role in batter performance, but this displayed little difference in bat speed and swing length (Figure 2). Similar results were observed for baserunner situations, innings, and other variables that were hypothesized as potentially impactful. Additionally, little to no correlations were found with quantitative pitcher measurements including velocity and movement. These results suggest that batters generally preserve similar tendencies independent of most factors, but there are some notable cases of adjustments.

One significant batter adjustment was changing swing length based on pitch location and pitch type. Figure 3 shows that the distributions are shifted by at least one quartile, indicating a major difference by pitch zone and pitch type. The explanation for pitch zone arises from the motion of a swing. Swings generally start from above the strike zone, so higher pitches require a shorter swing while lower pitches

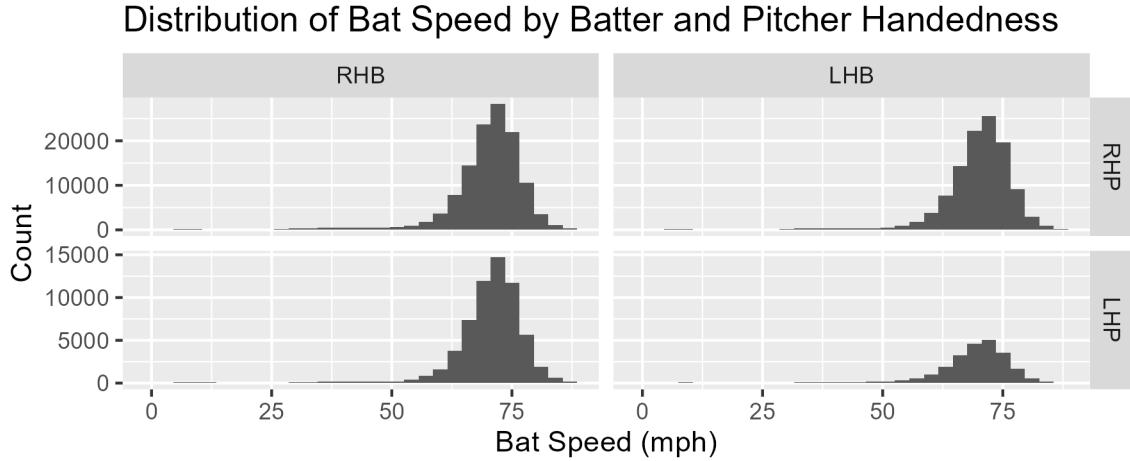


Figure 2: The distributions of bat speed are similar for each combination of batter and pitcher handedness.

require a longer swing. Higher velocity fastballs also require shorter, quicker swings than slower off-speed pitches. This finding generalizes the case study of Aaron Judge who has a drastically shorter swing on pitches up and in, and even Luis Arraez displays a similar, albeit weaker, pattern despite a more uniform contact-based approach (Petriello, 2024).

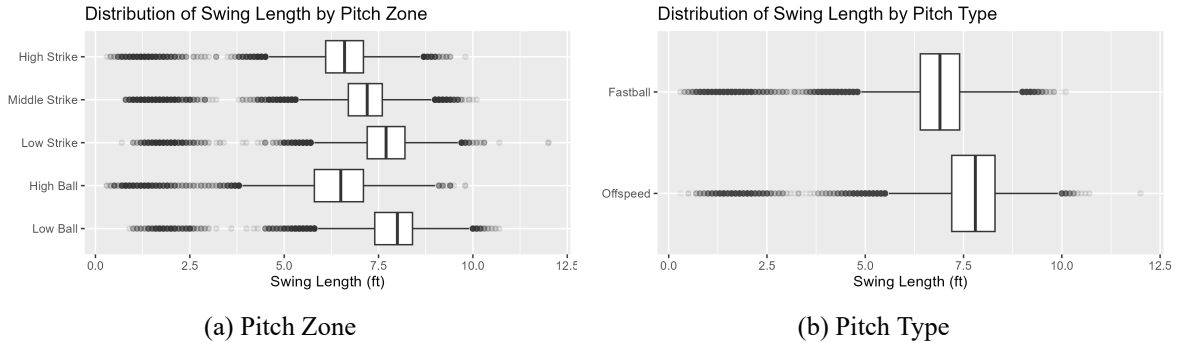


Figure 3: The distribution of swing length greatly differs by pitch location and pitch type. Higher pitches and fastballs correspond to shorter swings overall, while lower and offspeed pitches result in longer swings.

However, it is important to contextualize these differences using the relationship between pitch type and pitch zone. Figure 4 shows a significant difference in the percentage of fastballs thrown in each location, with almost no overlap in the middle 50% of each distribution. This approach to throw more high fastballs and low offspeed pitches agrees with conventional pitching wisdom and batting statistics, giving evidence that pitching tendencies similarly gravitate towards the optimum like batters. Observing this underlying relationship also reveals the mechanisms governing batter adjustments in Figure 3.

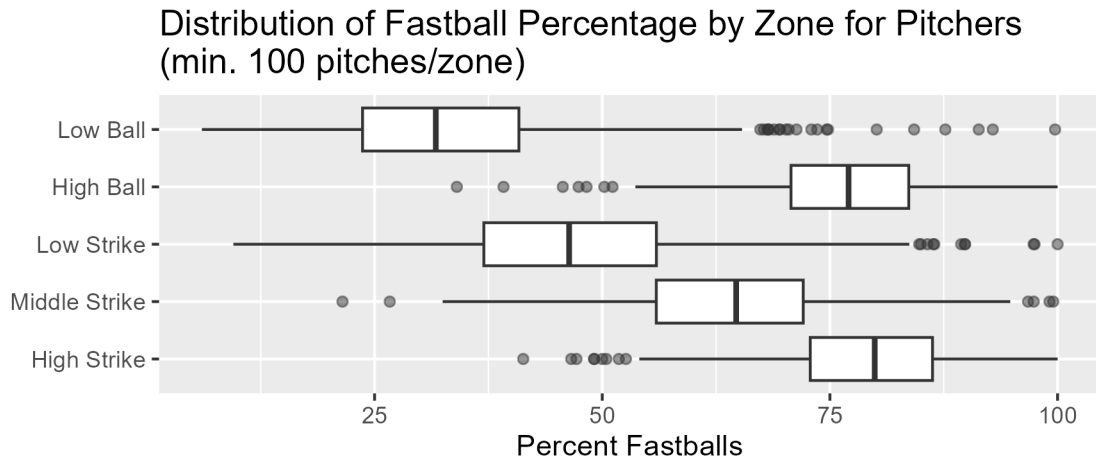


Figure 4: Pitchers throw a much greater percentage of fastballs in higher zones and lower percentage of fastballs in lower zones.

Another batter adjustment was changing bat speed by count, reducing bat speed in less favorable counts and increasing bat speed in more favorable counts. This shift is not as pronounced, but Figure 5 still shows a noticeable difference in the middle 50% of each distribution. This difference was not discussed in the initial analyses, but the explanation follows intuitively from a batter's intentions. In an unfavorable count, a slower bat speed increases the chance of making contact and avoiding a strikeout; in a favorable count, a faster bat speed aims to get a big hit, and there is less harm in swinging and missing. However, individual batters have different approaches to making this adjustment, measured by their fast-swing rate and swing distribution (Petriello, 2024). It is also interesting that this trend was less clear in swing length, reflecting a two-strike hitting approach through bat speed instead.

Bat Speed by Count (Balls and Strikes)

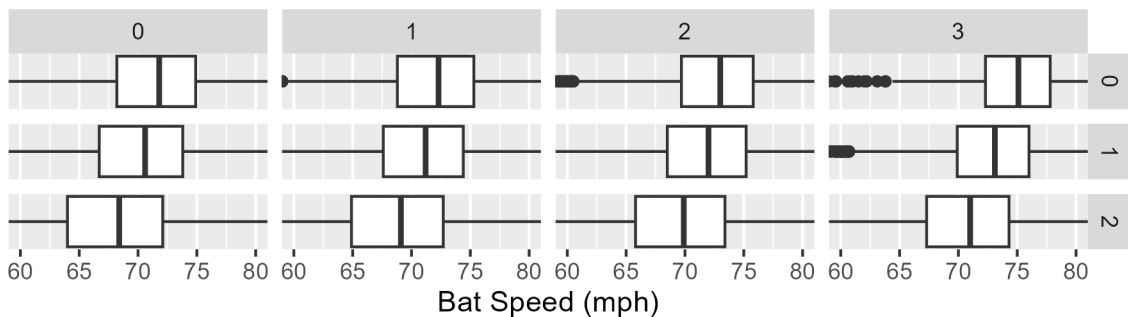


Figure 5: Batters slightly increase their bat speed in more favorable counts (top, right) and decrease bat speed in less favorable counts (bottom, left).

Another new but intuitive relationship was the moderate correlation observed between bat speed

and swing length (Figure 6). This can be understood through the physics of a swing: A longer swing requires a faster bat to cross home plate in the same time as a shorter swing. The correlation was slightly stronger (0.564) at the player level using median bat speed and swing length than at the swing level (0.528), but both exhibit a positive relationship. This analysis applies a systematic approach to understanding some case studies presented by Passan (2024). Players including Juan Soto and Freddie Freeman combine a fast bat speed with moderate to short swings, giving them the bat speed advantages while still using a high-contact approach. On the other hand, a long swing and slow bat speed could explain Nolan Arenado's offensive struggles. These outliers lend for interesting considerations, but the overall trend still governs most players.

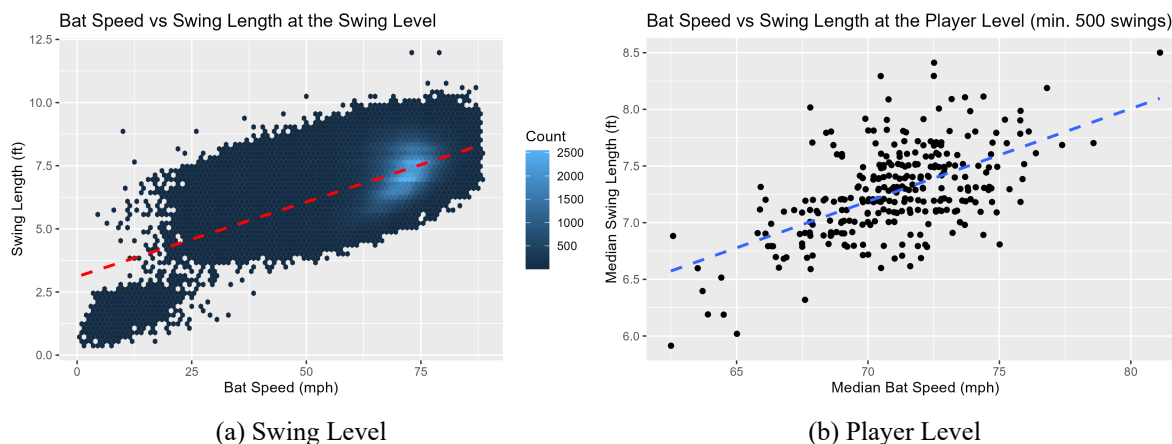


Figure 6: There is a moderate positive correlation between bat speed and swing length at both the swing (0.528) and player (0.564) level. Only players with at least 500 recorded swings were considered for the player level.

Relationship to Success

While understanding batter tendencies is interesting, the relationship of these qualities to success makes these results valuable. Petriello (2024) showed that faster swings significantly outperformed slower swings at the aggregate level, and this project seeks to consider the direct correlations. Success was quantified at the swing level using Statcast estimated batting average and wOBA value for each ball-in-play in the dataset, and a batter-level summary computed *pseudo*-batting average and wOBA metrics. These computations did not precisely match their definitions, but the values were comparable in representing a similar idea.

At the player level, bat speed and wOBA were moderately correlated with a correlation coefficient of 0.443 for 95th percentile bat speed and 0.429 for median bat speed. These relationships are plotted in Figure 7. The correlations between bat speed and batting average and swing length with both metrics were much weaker (<0.2). This agrees with the initial findings of bat speed as a better predictor of power, a metric captured in wOBA but not batting average, as well as the more stylistic implications of swing length (Petriello, 2024).

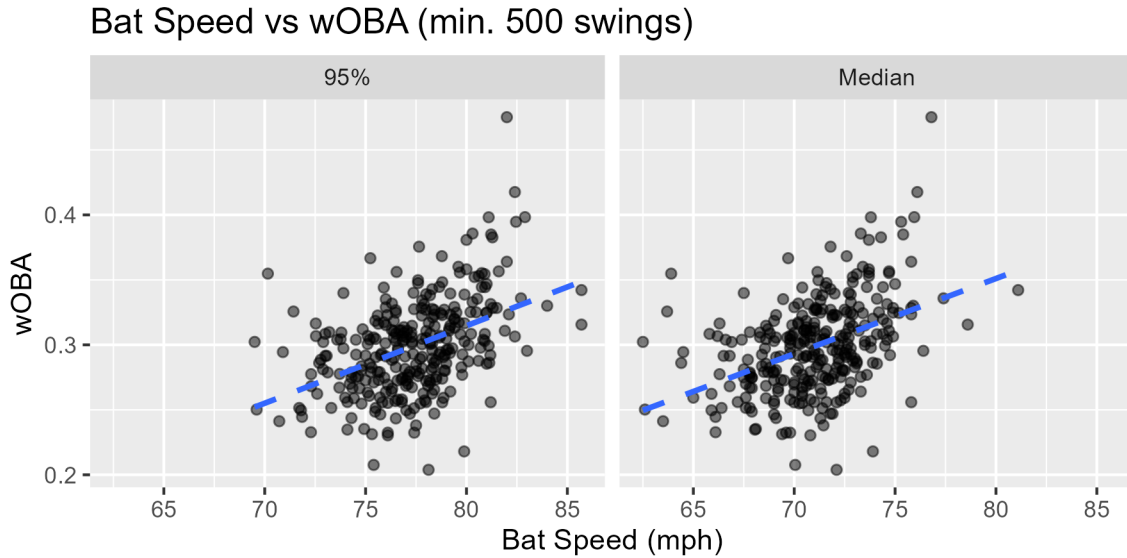


Figure 7: The strongest correlation at the player level exists between bat speed and wOBA. The correlation coefficient is 0.443 with the 95th percentile bat speed (left) and 0.429 with the median (right).

There was no notable correlation between either bat speed or swing length with success at the swing level. However, bat speed is clearly a meaningful statistic, and an indirect relationship is concealed by other factors which dictate a swing outcome. Figure 8 shows the moderate positive correlation (0.354) between bat speed and launch speed, reflecting the physical considerations of energy transfer. Launch speed is more closely related to success with moderate correlations to estimated batting average (0.377) and wOBA (0.290), making bat speed an important metric at the swing level despite having no direct correlation.

An interesting alternative perspective to consider the impact of bat speed is through the comparison between postseason and regular season games (Figure 9). While the distributions do not differ greatly, there is a noticeable shift towards faster bat speed in the postseason. This could simply be an artifact of sample sizes allowing a small subset of players with many postseason swings (including Stanton, Aaron Judge, and Shohei Ohtani) to skew the distribution. However, another related explanation is that faster bat speeds contributed to player success on postseason teams. Both factors likely play a role, and this shift could be interesting to revisit in future seasons if it persists.

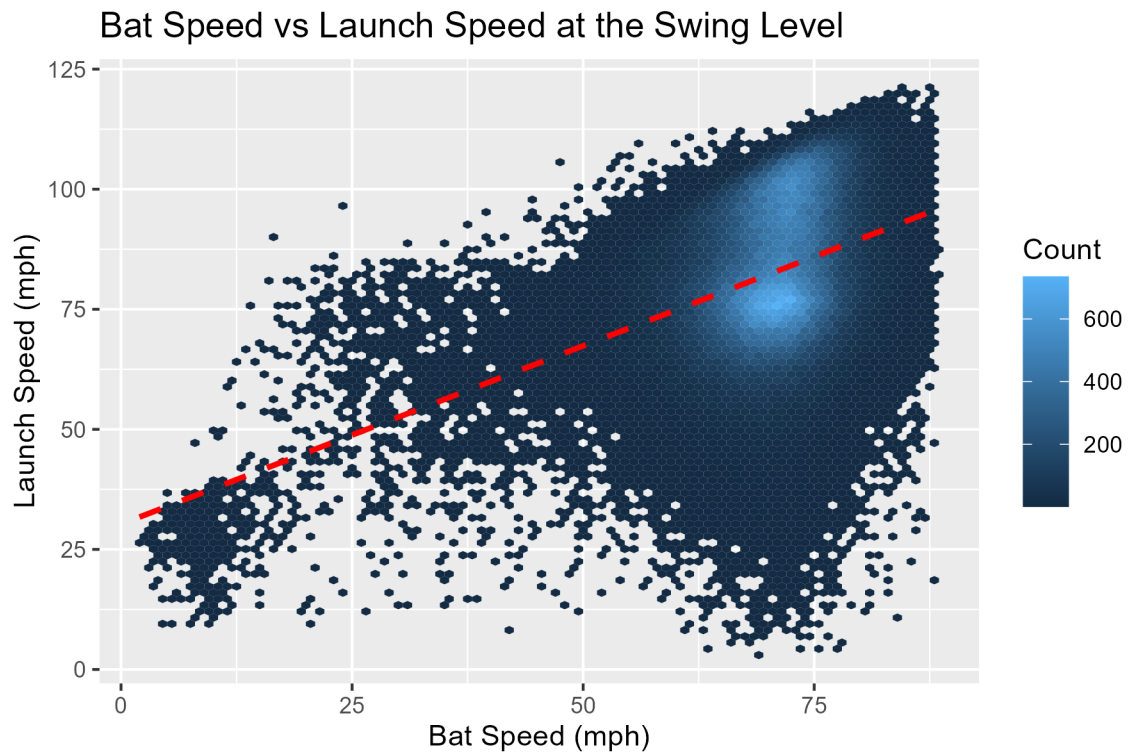


Figure 8: Bat speed is moderately correlated with launch speed at the swing level with a correlation coefficient of 0.354, illustrating an indirect relationship with success.

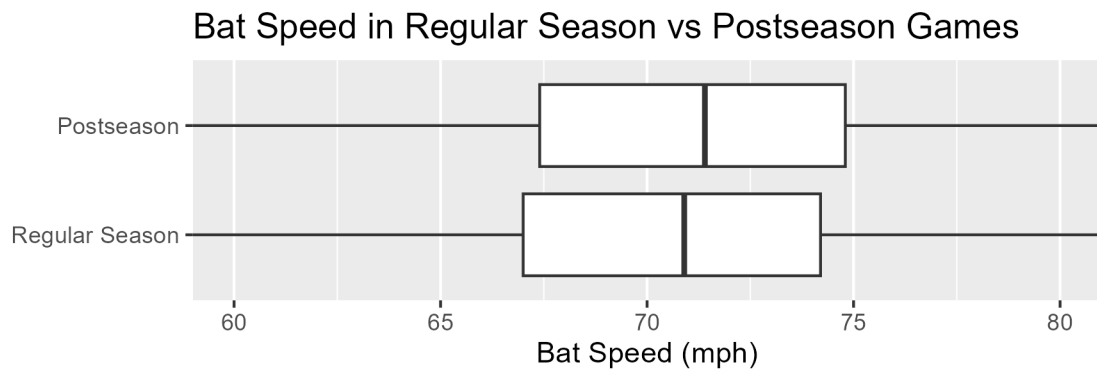


Figure 9: The distribution of bat speed shifts to the right slightly in postseason games compared to the regular season. This could be further evidence for bat speed as a predictor of success.

Discussion

While many potential relationships were hypothesized between batter tendencies and a range of situational and pitching variables, few displayed notable correlations. This reflects the limitations batters face with only milliseconds to identify and track a pitch from its release. With the observed adjustments based on pitch type and location, as well as the predetermined effects of count, there is little room for additional considerations, and thus other relationships were not found. In contrast, pitchers face no such constraint in making adjustments, so this understanding of batter tendencies may even be more beneficial for pitchers to strategize against a hitter. The current data is likely too limited to support this, but further research could make this a possibility in the near future.

The lack of any very strong correlations between bat speed and swing length with swing outcomes is also not surprising. The energy transfer of a batted ball is a complex phenomenon influenced by many other contact quality factors, and the actual outcome is further dependent on fielder positioning and luck. Many of these factors are nearly impossible to control, making hitting such a difficult skill that even the best players fail over 60 percent of the time. As such, the variable inputs of bat speed or swing length are highly unlikely to relate very closely to success. However, the moderate correlations give evidence for their role as additional factors to consider, and these metrics are analogous to spin rate and pitch movement as some of the first quantitative measurements for studying batters directly.

Overall, most of the results from this project did not differ greatly from expectations, and the consistency with the initial findings is notable. Some of the non-relationships were surprising, such as how baserunner situations and number of prior plate appearances hardly corresponded to changes in batter tendencies. These are variables that were expected to dictate a batter's approach, and one possible shortcoming of this project was in its sole focus on bat speed and swing length as such factors may instead be reflected in other qualities. Additionally, the limitation of considering correlations could also overlook other nonlinear relationships in the data. Nonetheless, these results are still very powerful in supporting bat speed and swing length as important metrics for studying batters.

Conclusion

The goal of this project was to understand natural batter swing tendencies with respect to bat speed and swing length, and to evaluate how these metrics related to on-field success. The results showed that batters generally do not adjust bat speed or swing length significantly, focusing primarily on pitch type and location for adjusting swing length and count for adjusting bat speed. An interesting relationship was also discovered in the positive correlation between bat speed and swing length, providing a useful tool for rationalizing changes in player performance. Finally, bat speed was found to be another important metric relating to batter success as previously hypothesized, though it is only one of many factors.

Baseball analytics is constantly evolving, trying to optimize every facet of the game, and the introduction of Statcast bat tracking data in the 2024 MLB Season opens many new possibilities for studying batters. One particular area of note is in the intersection with biomechanical data where these topics can be analyzed at a much smaller scale. For example, analyzing swing mechanics could reveal a slight adjustment in batting stance to achieve a greater maximum bat speed. The human aspect of baseball

will guarantee the presence of variance, but further research and changes in developmental programs could continue to reduce the amount of imperfection. Even without expanding to another field, simply analyzing additional bat tracking data from future seasons will also be valuable for generating new insights, such as how aging impacts bat speed. This project provided a starting point by confirming the importance of these metrics in characterizing batters, and there are many different directions for future research.

References

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